Andres Mosquera Hernandez

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Skills

- **Programming**: Python, Bash, C/C++/C#, Java, JavaScript, TypeScript, SQL, PHP, Kotlin, HTML5, CSS3, PostgreSQL.
- **AI/ML**: PyTorch, TensorFlow, Scikit-learn, LangChain, Hugging Face, Pandas, NumPy, OpenCV, Keras.
- **Cloud/Ops**: Docker, Kubernetes, AWS, Google Cloud (GCP), Azure, CI/CD, Git, GitHub, GitLab, Firebase, Linux.
- **AI Productivity**: ChatGPT, Cursor, GitHub Copilot, Codex, Claude Code, Vercel v0 (AI-assisted development).
- **Languages**: Spanish (Native), English (TOEFL 77, B2, Professional Working Proficiency).

Experience

- **Universidad de los Andes** — Bogota, Colombia
  Graduate Research Assistant (M.Sc.) — *Aug 2025 - Present*
  - **Research**: Research on Process Mining integrating Machine Learning, Reinforcement Learning, and LLMs for predictive and prescriptive process analytics. Design and evaluate ML/RL methods for runtime decision-making, intervention policies, and optimization.
  - **Publications**: Author and co-author of papers submitted to international conferences and journals in process mining, AI, and applied ML. Tools: Python, PyTorch, RL, Causal Inference, LLMs, Simulation, LaTeX.
  - **Service**: Poster Evaluator for IEEE-affiliated academic events, reviewing research posters by undergraduate and graduate students.

- **AI Safety Colombia** — Bogota, Colombia
  Facilitator, Technical AI Safety Seminar (BlueDot Impact curriculum) — *Oct 2026 - Nov 2026*
  - **Teaching**: Facilitating a 6-week technical AI safety seminar in Spanish: training safer models, evaluations, mechanistic interpretability, harm mitigation, and how to start contributing.

- **Universidad del Rosario** — Bogota, Colombia
  Researcher — *Jun 2025 - Jul 2025*
  - **AR Development**: Built AR environments in Unity for over 100 students to explore 3D bone models, using a virtual lab and auditorium (AR Foundation, Vuforia). Saved USD 2,000 in 3D printing costs via virtual inspection before physical production.

- **Coplanin** — Bogota, Colombia (Remote)
  Machine Learning Engineer / Full Stack / Android Engineer — *April 2023 - June 2025*
  - **Machine Learning**: Trained a Pix2Pix GAN reaching 0.87 IoU and 92% accuracy in cadastral image segmentation, cutting manual rework by 60%. Fine-tuned Gemma-2, Llama 3, Mistral, Phi-4, and Qwen2.5 to split 57,000 Spanish names (81% accuracy) and infer gender (95%). Classified 35,000 addresses at 93% accuracy. Stack: PyTorch, Hugging Face, Unsloth, OpenCV, PostgreSQL.
  - **Full Stack**: ETL from shapefiles to territorial objects for the Colombian Ministry of Environment via Airflow (3x faster than manual). Built ColSmart IGAC for property management (React, Next.js, PostgreSQL, TypeScript, Mapbox, QGIS) and contract systems for the Government of Cundinamarca (Java, Spring Boot, jQuery, AJAX).
  - **Mobile**: Built property-management and water-network apps (Android, Jetpack Compose, Kotlin, Mapbox, ArcGIS), one used by 700 users. Led two engineers to deliver 2x faster.

- **Kohl's** — Remote
  Machine Learning Ops Intern — *Aug 2024 - Oct 2024*
  - **MLOps**: Automated deployment of AI models on datasets with 10 million records for a retail U.S. company (96,000 employees), reducing deployment time by 73%. Stack: Python, Kubernetes, Vertex AI, Airflow, Google Cloud, Cloud Composer, Docker.

- **Alpha Intelligence AI** — Bogota, Colombia
  Software Engineer Intern — *Oct 2022 - Jan 2023*
  - **Software**: Analyzed data from Skaphe's 10-terabyte database for the government's Judicial Branch project (Python, SQL, Azure). Built the company website (JavaScript, jQuery, HTML5, CSS), increasing visibility by 25%.

- **Corparques** — Bogota, Colombia
  Software Engineer Intern — *May 2022 - July 2022*
  - **Software**: Built a search engine for the Mundo Aventura website (JavaScript, jQuery), used by 50 employees. Automated operational tasks, reducing completion time by 87% (Python, Power Automate).

Education

- **Universidad de los Andes** — Bogota, Colombia
  M.Sc. in Systems and Computing Engineering — *Aug 2025 – June 2027*
  - **Awards**: Full scholarship (Beca Avanza) as a research assistant.
  - **Focus**: Artificial Intelligence, Machine Learning, Natural Language Processing, Systems Engineering.
- **BlueDot Impact** — Remote
  Technical AI Safety Course and Technical AI Safety Project — *2026*
  - **Focus**: Alignment, mechanistic interpretability, evaluations, AI control, and scalable oversight. Also completed the Future of AI course.
  - **Project**: Technical AI Safety Project mentored by Jess Bergs (UK AI Security Institute).
- **Uniempresarial – Chamber of Commerce of Bogota** — Bogota, Colombia
  B.Sc. in Software Engineering, summa cum laude; GPA: 4.6/5 — *Aug 2021 – March 2025*
  - **Awards**: Ranked 1st in graduation class. 7x Best Course Programming Project. Highest Saber Pro score (Colombia's national standardized exam) in the program.
  - **Coursework**: Data Structures, Algorithms, AI, Computer Architecture, Distributed Systems, OOP.
  - **Role**: AI Teaching Assistant. Conducted AI research at the National University of Colombia; developed PhingPT, a model specialized for the financial sector.

Selected Publications

- **Thought Anchors for Social Bias: Which Reasoning Steps Matter in Extended Thinking LLMs on Latin American Scenarios**: Global South AI Safety Hackathon (Apart Research), 2026 – Winner (1 of 3 winning teams out of 217; solo). Adapted the Thought Anchors resampling framework to chain-of-thought interpretability, attributing which reasoning steps shift extended-thinking LLMs (Claude Sonnet 4.6, Gemini 2.5 Flash, Qwen3.7-Plus) toward stereotypical outputs. Shipped a reproducible pipeline, paper, and interactive demo in ∼24 hours. **A. Mosquera-Hernandez**.
- **When One Step Is Not Enough**: International Conference on Business Process Management (BPM 2026). Hybrid extractive–abstractive LLM pipelines (LangChain/LlamaIndex), multi-agent architectures, and evaluation (ROUGE-2, ROUGE-L, BLEU). 3rd place in the JUST-NLP 2025 Shared Task. **A. Mosquera-Hernandez** et al.
- **Automatic Legal Judgment Summarization Using Large Language Models**: Under review for journal publication. Case study for the JUST-NLP 2025 Shared Task; LLM pipelines for abstractive and reference-based summarization, dataset curation, and evaluation. **A. Mosquera-Hernandez** et al.

Conferences/Awards

- **Winner – Global South AI Safety Hackathon (Apart Research, 2026)**: 1 of 3 winning teams out of 217, competing solo. USD 1,000 prize.
- **Speaker & Poster Presenter – International Symposium of Statistics (Pasto, Colombia, 2025)**: Financial Sentiment Analysis with Large Language Models (RAG, fine-tuning).
- **1st Place – AI Learning Tools Challenge by Scotiabank Colpatria (2024)**: Among 32 universities. Stack: React, Next.js.
- **Applied Projects (2024)**: WhatsApp Chatbot for Cusezar (JavaScript, Salesforce, WhatsApp Business API); Task Manager for Colmenares Law Firm (AngularJS, Node.js, Docker, MySQL, GCP).